



4.5 Discrete distributions and the binomial distribution

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A discrete random variable X has a probability distribution $P(X=x)$, with $E(X)=\sum x \cdot P(X=x)$ and $\text{Var}(X)=E(X^2)-[E(X)]^2$. The binomial distribution $X \sim B(n,p)$ models the number of successes in n independent trials, each with constant success probability p : $P(X=x)=C(n,x)p^x(1-p)^{n-x}$, with $E(X)=np$ and $\text{Var}(X)=np(1-p)$. Probabilities are usually computed using technology (binomial pdf/cdf functions).

Worked example: follow the reasoning

A multiple-choice quiz has 10 questions, each with 4 options and 1 correct answer. A student guesses randomly on every question, so $X \sim B(10,0.25)$. Find (a) $P(X=3)$, (b) the expected number of correct answers.

Solution: (a) $P(X=3)=C(10,3)(0.25)^3(0.75)^7=120 \times 0.015625 \times 0.13348 \approx 0.250$. (b) $E(X)=np=10 \times 0.25=2.5$.

Quick check · multiple choice

$X \sim B(6,0.5)$. What is $E(X)$?

- A 2
- B 3
- C 4
- D 6

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A factory finds that 8% of the light bulbs it produces are defective. A random sample of 15 bulbs is taken, and X , the number of defective bulbs, satisfies $X \sim B(15,0.08)$. (a) Find $P(X=2)$, to 3 significant figures. (b) Find $P(X \leq 1)$, to 3 significant figures. (c) Find the expected number and standard deviation of defective bulbs in the sample, to 3 significant figures. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $P(X=2)=C(15,2)(0.08)^2(0.92)^{13}=105 \times 0.0064 \times 0.3383 \approx 0.227$. (b) $P(X=0)=0.92^{15} \approx 0.286$; $P(X=1)=15 \times 0.08 \times 0.92^{14} \approx 0.373$; $P(X \leq 1) \approx 0.286 + 0.373 = 0.660$. (c) $E(X)=np=15 \times 0.08=1.2$; $\text{Var}(X)=np(1-p)=15 \times 0.08 \times 0.92=1.104$; $SD=\sqrt{1.104} \approx 1.05$.

Quick check answer

B. $E(X)=np=6 \times 0.5=3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

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